

A Hyper parameter search

In this appendix, we present F1 scores during cross-validation of ELMoAR and GPTAR with, respectively, different window sizes and different head and layer numbers. The best hyperparameters are selected for the ELMoHAR-note, ELMoHAR and GPTHAR-note and GPTHAR algorithms.

For ELMoAR, the best hyperparameters are: a window size of 60. For GPTAR, the best hyperparameters are : 8 attention heads and 3 layers of decoder.

Table 7: Cross-validation F1-Scores with their standard deviation for GPTAR and ELMoAR models with various hyperparameters across the three datasets, to assess the Impact of context window size in ELMoAR and the number of layers and attention heads in GPTAR

	Aruba		Milan		Cairo		Average
	Macro F1 Score	std	Macro F1 Score	std	Macro F1 Score	std	
ELMoAR (Window 20)	83.47	1.83	71.10	2.25	66.37	3.24	73.65
ELMoAR (Window 40)	82.93	2.12	70.73	2.43	66.70	4.07	73.45
ELMoAR (Window 60)	83.47	2.61	72.40	2.49	67.23	4.70	74.37
GPTAR (8 Heads 3 Layers)	83.20	1.45	73.77	2.19	70.90	3.48	75.96
GPTAR (8 Heads 4 Layers)	83.53	1.48	72.30	1.62	69.03	4.42	74.95
GPTAR (12 Heads 6 Layers)	83.57	1.57	73.07	2.13	68.27	4.08	74.97

B Detailed Algorithms Metrics

In this appendix, we present a comprehensive breakdown of the performance metrics for the algorithms used in our study. The table encompasses test results from three distinct datasets: Aruba (Table 8), Milan (Table 9), and Cairo (Table 10). These metrics show that for the most simple dataset, Aruba, ELMoHAR and GPTHAR perform closely, ranking first or second depending on the chosen metric. For the more complex datasets, Milan and Cairo, GPTHAR outperforms all the other algorithms, regardless of the choice of metric.

Table 8: Detailed Algorithms Scores over the datasets Aruba

	ELMoAR	GPTAR	ELMoHAR-note	GPTHAR-note	ELMoHAR	GPTHAR
Accuracy	97.00%	97.10%	98.40%	98.20%	98.50%	98.52%
Precision	86.00%	90.30%	89.90%	87.40%	94.30%	91.20%
Recall	84.70%	85.10%	88.10%	88.60%	89.70%	89.80%
F1 Score	84.80%	86.10%	88.10%	87.30%	90.70%	89.70%
Balanced Accuracy	84.76%	85.18%	88.22%	88.55	89.71%	90.10%
Weighted Precision	96.70%	97.00%	98.10%	98.00%	98.50%	98.00%
Weighted Recall	97.00%	97.00%	98.40%	92.20%	98.50%	98.30%
Weighted F1 Score	96.9%	97.00%	98.00%	98.00%	98.30%	98.20%

Table 9: Detailed Algorithms Scores over the dataset Milan

	ELMoAR	GPTAR	ELMoHAR-note	GPTHAR-note	ELMoHAR	GPTHAR
Accuracy	87.50%	88.20%	90.00%	91.30%	90.60%	91.9%
Precision	75.90%	80.20%	85.60%	87.60%	88.90%	90.00%
Recall	68.40	68.50%	73.90	76.90%	75.60%	79.20%
F1 Score	70.80%	70.80%	77.40%	79.90%	80.00%	81.90%
Balanced Accuracy	68.51%	68.55%	73.91%	76.87%	77.84%	79.22%
Weighted Precision	86.80%	88.20%	89.60%	91.20%	90.70%	91.90%
Weighted Recall	87.50%	88.20%	90.00%	91.30%	90.60%	91.90%
Weighted F1 Score	86.70%	87.60%	89.20%	90.70%	90.00%	91.70%

Table 10: Detailed Algorithms Scores over the dataset Cairo

	ELMoAR	GPTAR	ELMoHAR-note	GPTHAR-note	ELMoHAR	GPTHAR
Accuracy	81.1%	83.40%	87.30%	91.00%	90.80%	93.20
Precision	72.70%	76.60%	79.70%	87.40%	87.30%	89.80%
Recall	69.20%	71.40%	74.70%	83.40%	82.00%	86.60%
F1 Score	70.50%	73.20%	75.90%	84.80%	83.30%	87.20%
Balanced Accuracy	69.12%	71.33%	74.75%	83.58%	81.87%	86.74%
Weighted Precision	81.10%	82.40%	86.90%	91.10%	90.60%	93.20
Weighted Recall	81.10%	83.40%	87.30%	91.00%	90.80%	93.20
Weighted F1 Score	80.90%	82.30%	86.80%	90.50%	90.30%	92.70%

C Confusion Matrix

In this section, we report on the confusion matrices for various algorithms across three datasets. A notable observation is that the more complex architectures, which include time encodings (namely ELMoHAR and GPTHAR), exhibit fewer misclassifications compared to simpler models.

This improvement is particularly evident in specific activities. For instance, in the Aruba dataset, activities like 'Washing Dishes', 'Meal Preparation', 'Enter Home', and 'Leave Home' are classified more accurately. Similarly, in the Milan dataset, activities such as 'Eve Med' and 'Morning Meds' show a marked decrease in misclassifications. Additionally, in the Cairo dataset, meal-related activities like 'Breakfast', 'Lunch', and 'Dinner' are more accurately identified.

These findings highlight the effectiveness of our improved algorithms in reducing misclassifications.

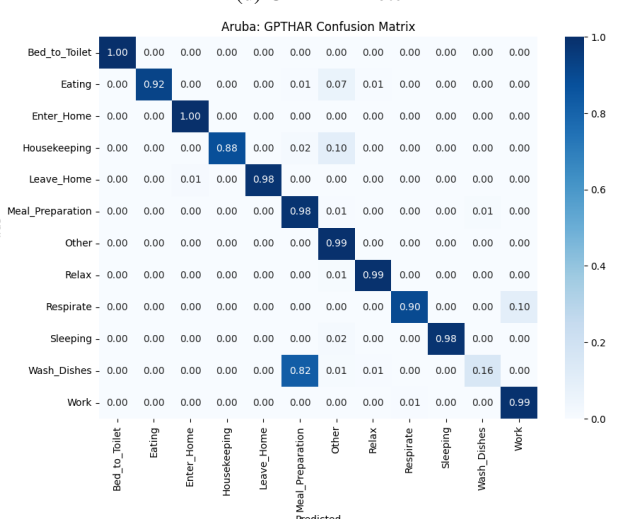
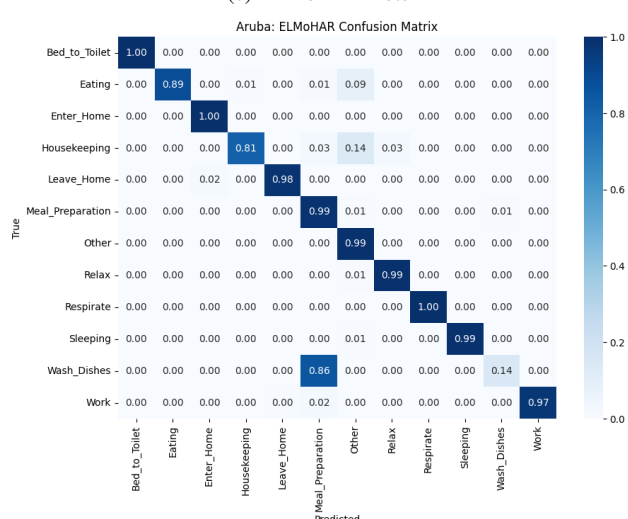
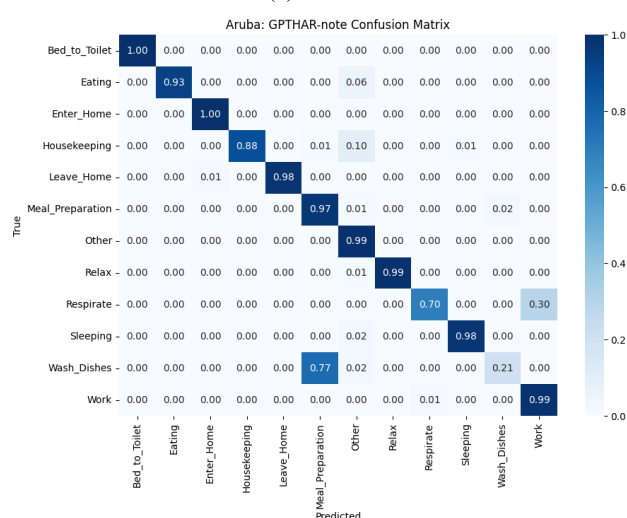
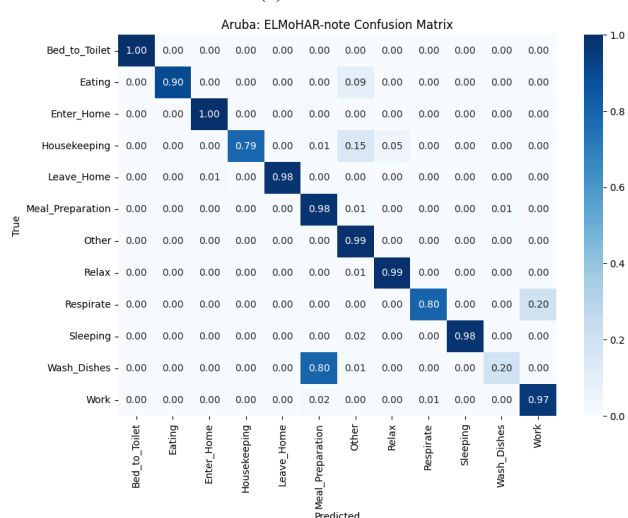
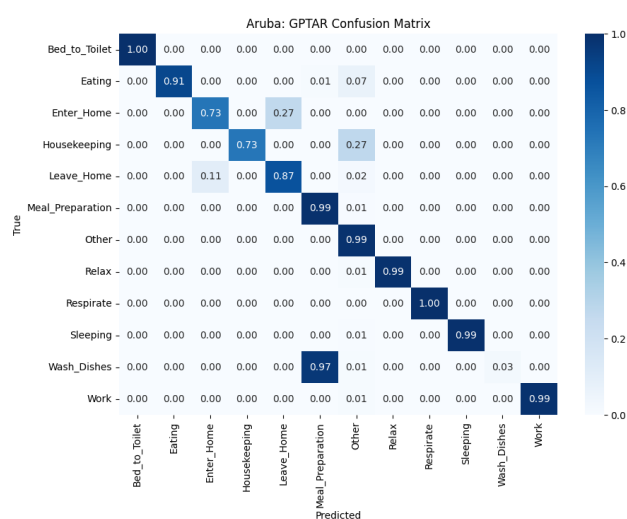
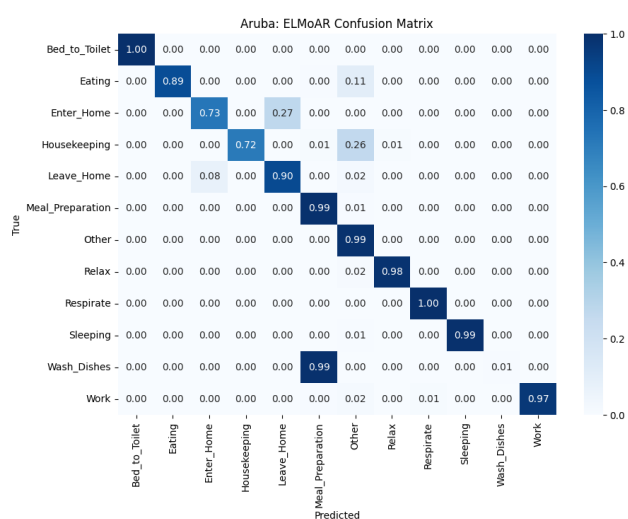
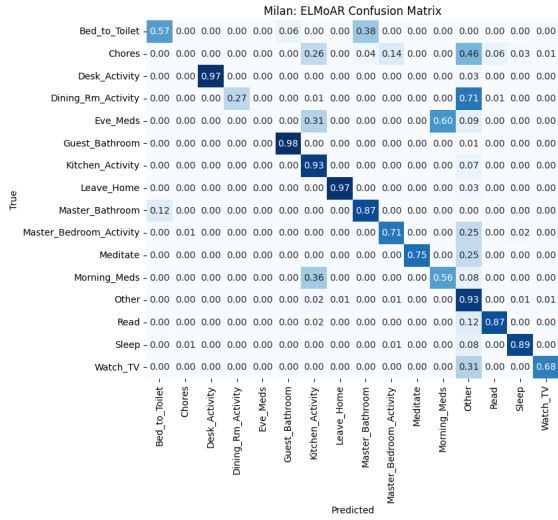
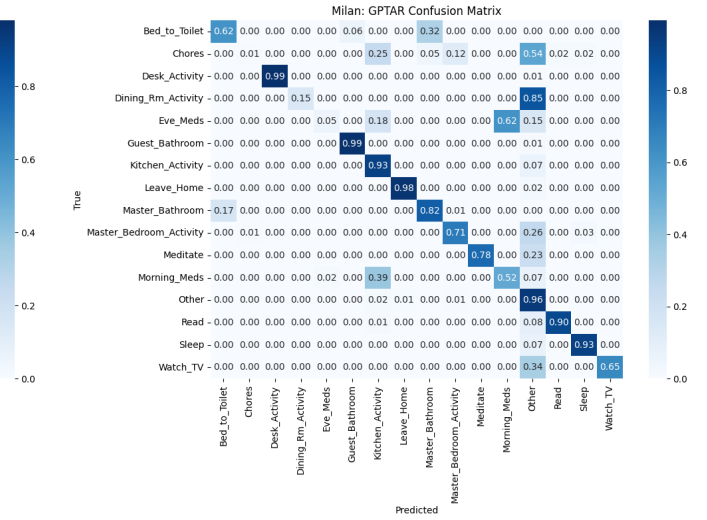


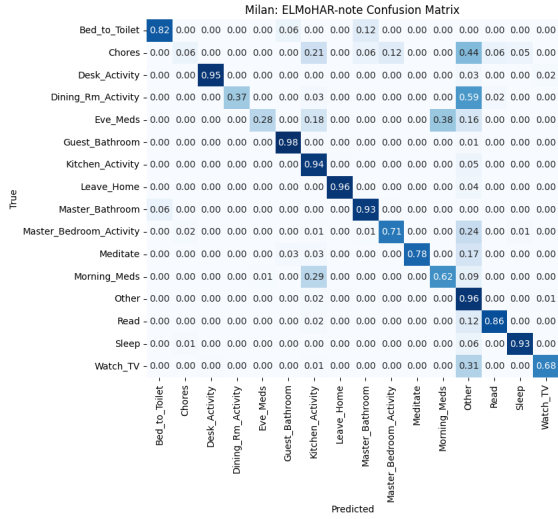
Figure 7: Confusion matrices per algorithm on the Aruba dataset.



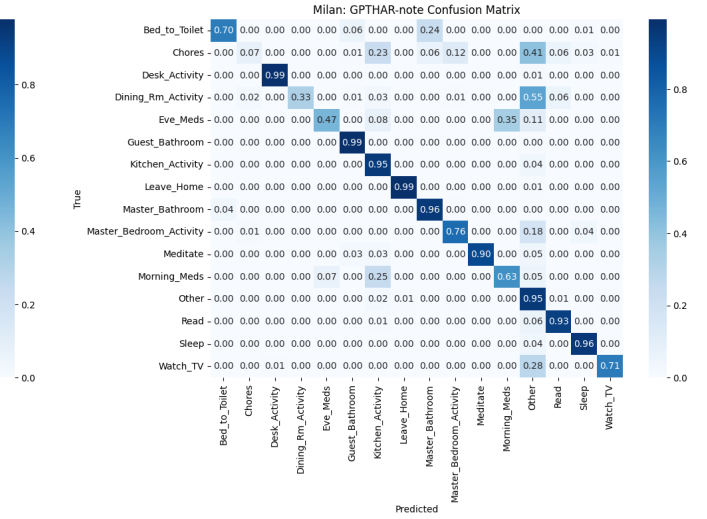
(a) ELMoAR



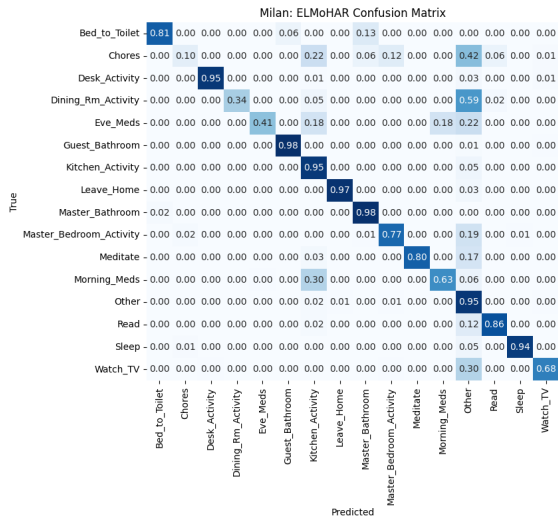
(b) GPTAR



(c) ELMoHAR-note



(d) GPTHAR-note



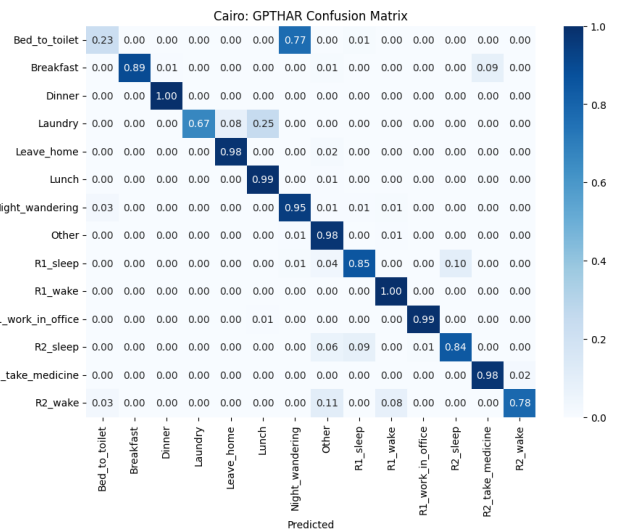
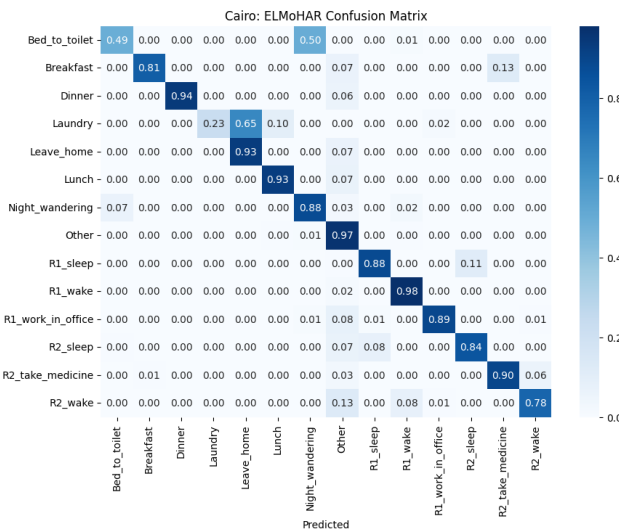
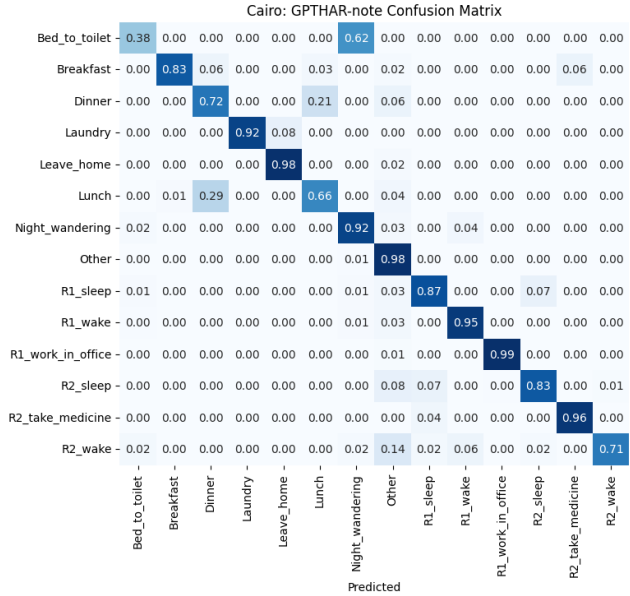
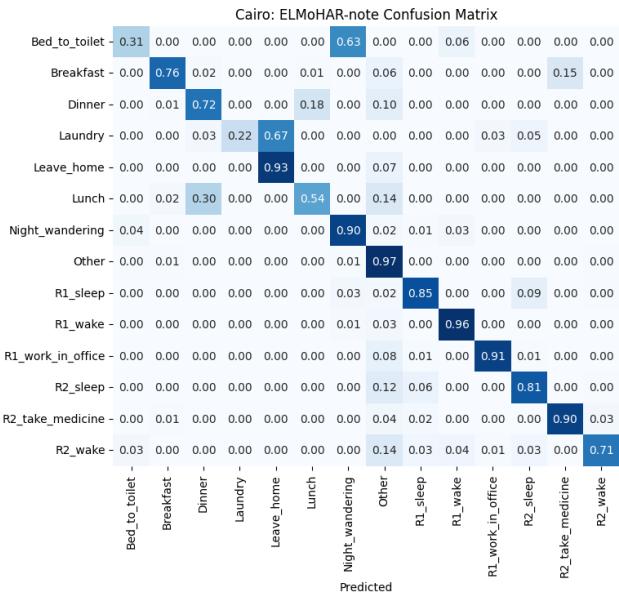
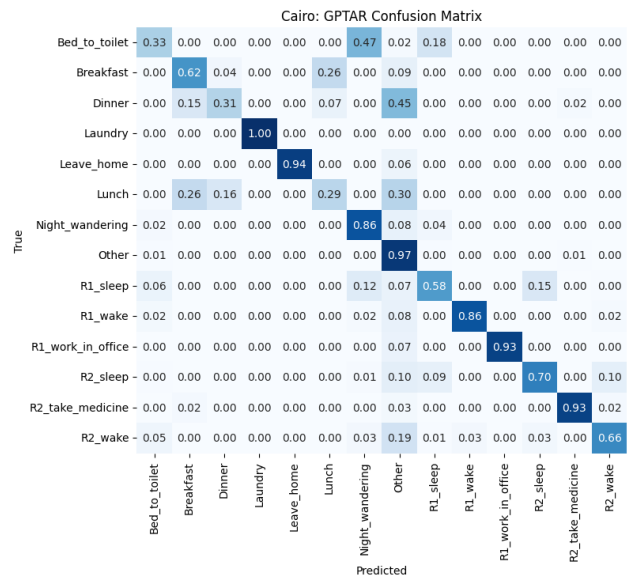
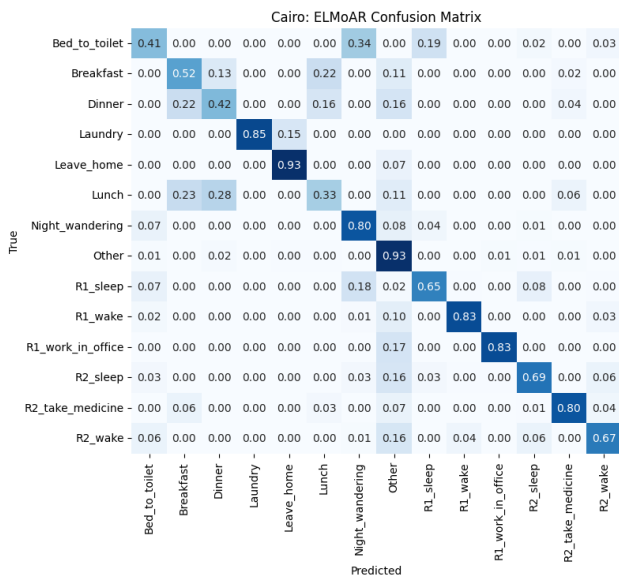


Figure 9: Confusion matrices per algorithm on the Cairo dataset.

D Datasets Details

This appendix details information on three CASAS datasets—Aruba, Milan, and Cairo—created by Washington State University [5]. Data were collected from real homes equipped with smart sensors and inhabited by real residents. The sensor events log, as shown in Figure 10, comprises motion sensors (sensor IDs beginning with "M"), door closure sensors (sensor IDs beginning with "D"), and temperature sensors (sensor IDs beginning with "T"). Each log entry consists of a single event per row and includes six columns: date, timestamp, sensor ID, sensor value, activity during which the sensor was triggered, and the activity status (its beginning or its end). The datasets differ in housing structure, number of inhabitants, and the duration of activity labeling, with several months of data for each (details provided in Table 11). Floor maps for each dataset are illustrated in Figure 11.

Aruba and Milan feature single-floor homes with the same number of rooms but different layouts, whereas the Cairo home spans three floors and contains more rooms than the former two. The Aruba dataset captures the daily activities of a woman living alone, Milan documents the life of a woman and her dog, and Cairo describes a couple living with a dog.

The three datasets are unbalanced, with some activities less represented than others, as depicted in Table 12. This imbalance is due to the real-life settings of these homes and the lifestyles of the inhabitants. Furthermore the datasets do not cover exactly the same activities

2010-11-04	05:40:51.303739	M004	ON	Bed_to_Toilet	begin
2010-11-04	05:40:52.342105	M005	OFF		
2010-11-04	05:40:57.176409	M007	OFF		
2010-11-04	05:40:57.941486	M004	OFF		
2010-11-04	05:43:24.021475	M004	ON		
2010-11-04	05:43:26.273181	M004	OFF		
2010-11-04	05:43:26.345503	M007	ON		
2010-11-04	05:43:26.793102	M004	ON		
2010-11-04	05:43:27.195347	M007	OFF		
2010-11-04	05:43:27.787437	M007	ON		
2010-11-04	05:43:29.711796	M005	ON		
2010-11-04	05:43:30.279021	M004	OFF	Bed_to_Toilet	end
2010-11-04	05:43:45.7324	M003	ON	Sleeping	begin
2010-11-04	05:43:52.044085	M003	OFF		
2010-11-04	05:43:53.185335	M002	ON		
2010-11-04	05:43:53.253809	M003	ON		
2010-11-04	05:43:59.493281	M002	OFF		

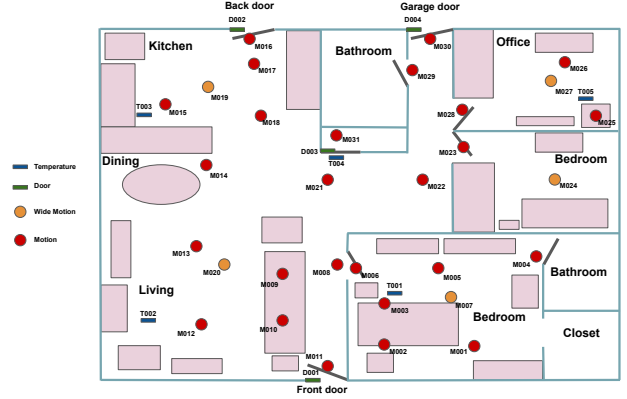
Figure 10: Extract of a CASAS dataset log

Table 11: Details of the three CASAS datasets

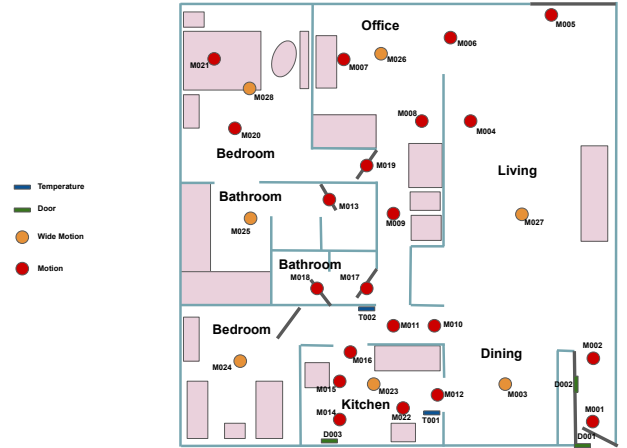
Dataset	Aruba	Milan	Cairo
Residents	1	1 + pet	2 + pet
Number of Sensors	39	33	27
Number of Activities	12	16	13
Number of Days	219	82	56

Table 12: Aruba, Milan and Cairo labels details.

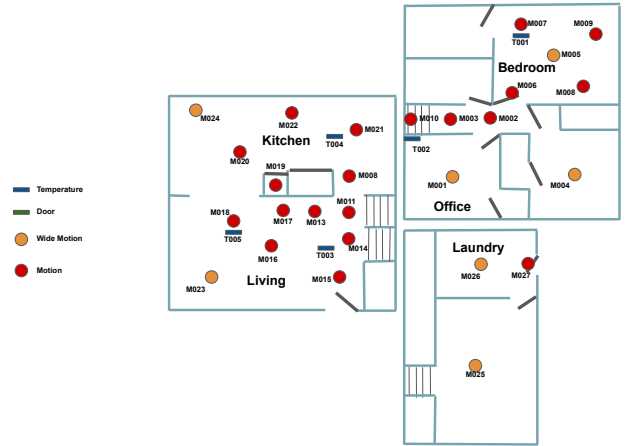
Aruba		Milan		Cairo	
Labels	# occurrence	Labels	# occurrence	Labels	# occurrence
Bed_to_Toilet	155	Bed_to_Toilet	86	Bed_to_toilet	29
Eating	262	Chores	32	Breakfast	49
Enter_Home	427	Desk_Activity	54	Dinner	42
Housekeeping	34	Dining_Rm_Activity	26	Laundry	10
Leave_Home	427	Eve_Meds	19	Leave_home	69
Meal_Preparation	1610	Guest_Bathroom	335	Lunch	37
Other	6079	Kitchen_Activity	741	Night_wandering	66
Relax	2944	Leave_Home	211	Other	579
Respirate	6	Master_Bathroom	307	R1_sleep	55
Sleeping	416	Master_Bedroom_Activity	204	R1_wake	53
Wash_Dishes	64	Meditate	17	R1_work_in_office	46
Work	171	Morning_Meds	41	R2_sleep	53
		Other	1776	R2_take_medicine	44
		Read	368	R2_wake	64
		Sleep	166		
		Watch_TV	153		



(a) Aruba dataset floor map



(b) Milan dataset floor map



(c) Cairo dataset floor map

Figure 11: Floor maps of Aruba, Milan, and Cairo Datasets.

E Experiments Reproduction

This appendix presents the scores and results of our replication of the Liciotti et al. [13] experiments. Table 13 below includes results from the original paper for the Milan and Cairo datasets, alongside our findings. It’s important to note that the Aruba dataset was not explored in the original study. We conducted our experiments 10 times, and the table reflects the average results of these repetitions. In line with the original paper, we employed a 3-fold cross-validation evaluation method for each dataset and regrouping the original dataset labels under meta-activities using the same activity remapping as in the original study. Additionally, we adhered to the same hyperparameters defined in the original paper to ensure consistency in our replication process. Our findings demonstrate that we achieved results closely similar to those in the original paper, confirming the accuracy of our implementation of the algorithm.

Table 13: Reproduction results of the Bi-LSTM architecture as proposed by Liciotti et al. Our replication of these experiments was conducted 10 times to ensure reliability and consistency of the results.

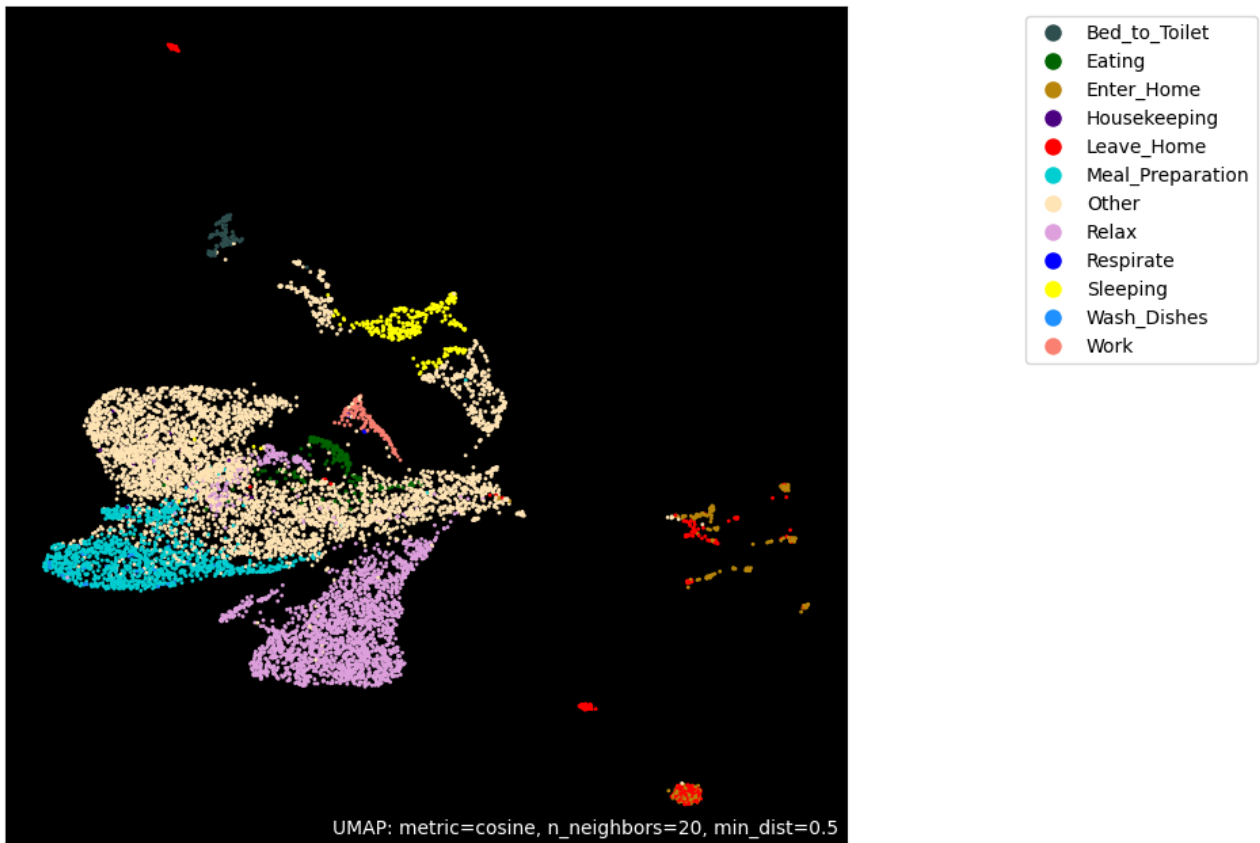
	Aruba		Milan		Cairo	
	Liciotti et al. (paper)	Liciotti et al. (reproduce)	Liciotti et al. (paper)	Liciotti et al. (reproduce)	Liciotti et al. (paper)	Liciotti et al. (reproduce)
Accuracy	NA	96.17%	94.12%	90.70%	86.90%	86.67%
Precision	NA	92.73%	NA	82.33%	NA	79.67%
Recall	NA	90.17%	NA	77.00%	NA	75.00%
F1 Score	NA	91.17%	NA	79.33%	NA	76.33%
Balanced Accuracy	NA	90.37%	NA	76.88%	NA	74.98%
Weighted Precision	NA	96.30%	94.00%	90.33%	86.67%	86.33%
Weighted Recall	NA	96.10%	94.00%	90.67%	87.00%	86.67%
Weighted F1 Score	NA	96.03%	94.00%	90.33%	86.67%	86.33%

F Activities Embedding

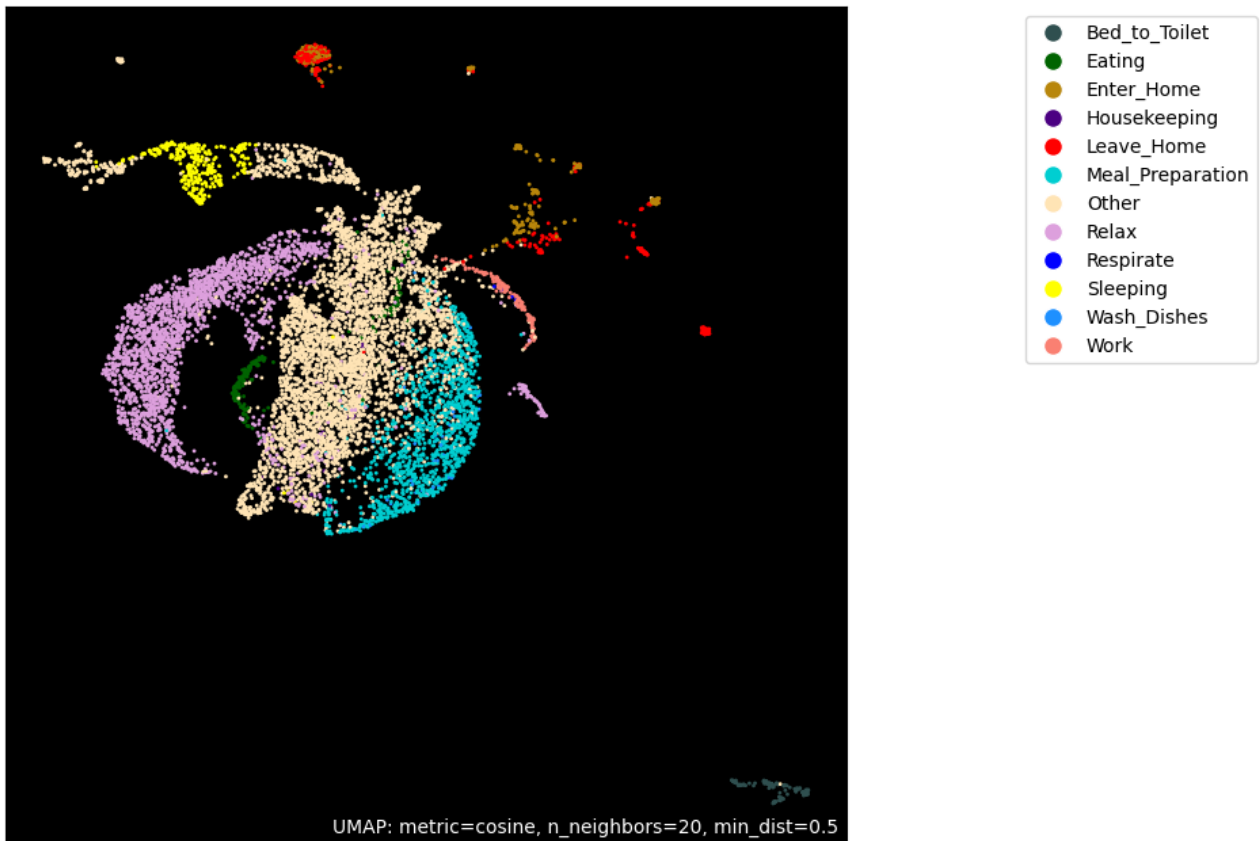
In this appendix, we present the embeddings of activities from three datasets: Aruba, Milan, and Cairo. These embeddings effectively capture the relationships between various activities and provide a visual representation of the feature space for each activity. We conduct a comparative analysis of the embeddings generated by the ELMo and GPT models. The left images in our figures showcase activities embedded using the ELMo models, while the right images illustrate those embedded by the GPT models.

To generate the embedding of activity sequences, each pre-segmented activity from the datasets is processed through the respective embedding model. The resulting output vectors are then averaged to produce a unique vector representation for each activity. Subsequently, these vectors are dimensionally reduced using the Uniform Manifold Approximation and Projection (UMAP) algorithm, enabling their representation in a 2-dimensional space. Each point in this space is color-coded according to its corresponding activity label. Figures 12, 13, and 14 demonstrate the ELMo-based and GPT-based activity sequence embeddings for the Aruba, Milan, and Cairo datasets, respectively.

We observe that the embeddings are generally similar. However, it is crucial to note that the GPT embedding is trained from random chunks that may not contain only one activity but potentially a sub-part or parts of multiple activities in contrary of ELMo which use pre-segmented activities. We can observe, in the Milan dataset, the GPT embedding appears to more distinctly separate some clusters, for instance, it isolate the ‘master_bathroom’ activities. For the Cairo dataset, the GPT embedding seems to segregate all activities into two main clusters: one grouping day activities and the other night activities.

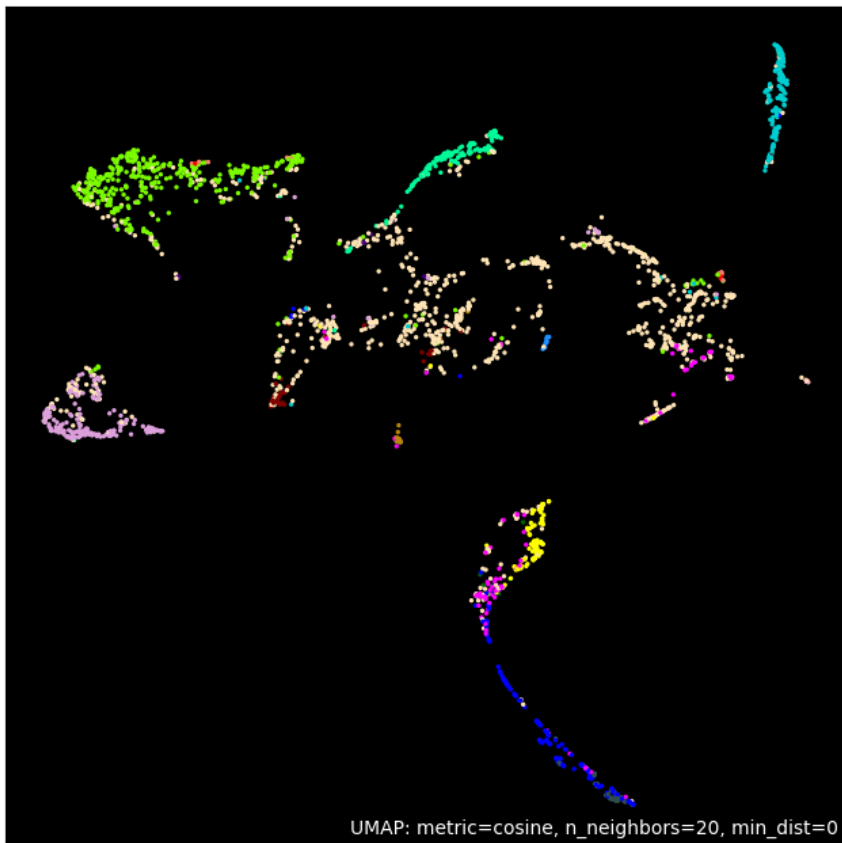


(a) ELMo



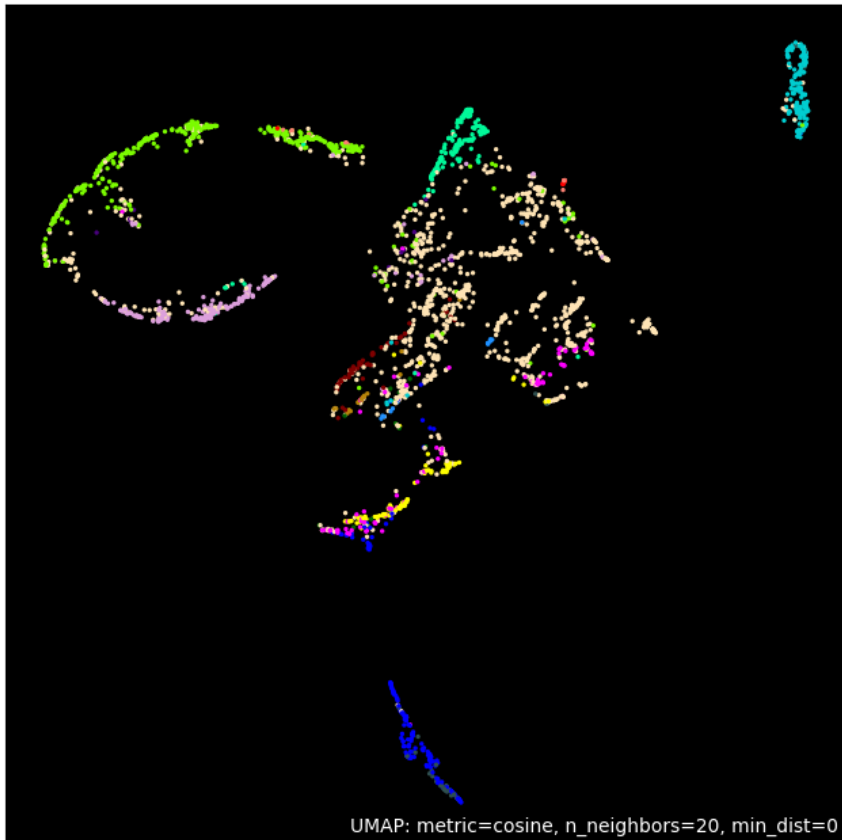
(b) GPT

Figure 12: Visualization of Activity Embeddings for the Aruba Dataset (Test Set) Generated by ELMo (Top) and GPT (Bottom)



- Bed_to_Toilet
- Chores
- Desk_Activity
- Dining_Rm_Activity
- Eve_Meds
- Guest_Bathroom
- Kitchen_Activity
- Leave_Home
- Master_Bathroom
- Master_Bedroom_Activity
- Meditate
- Morning_Meds
- Other
- Read
- Sleep
- Watch_TV

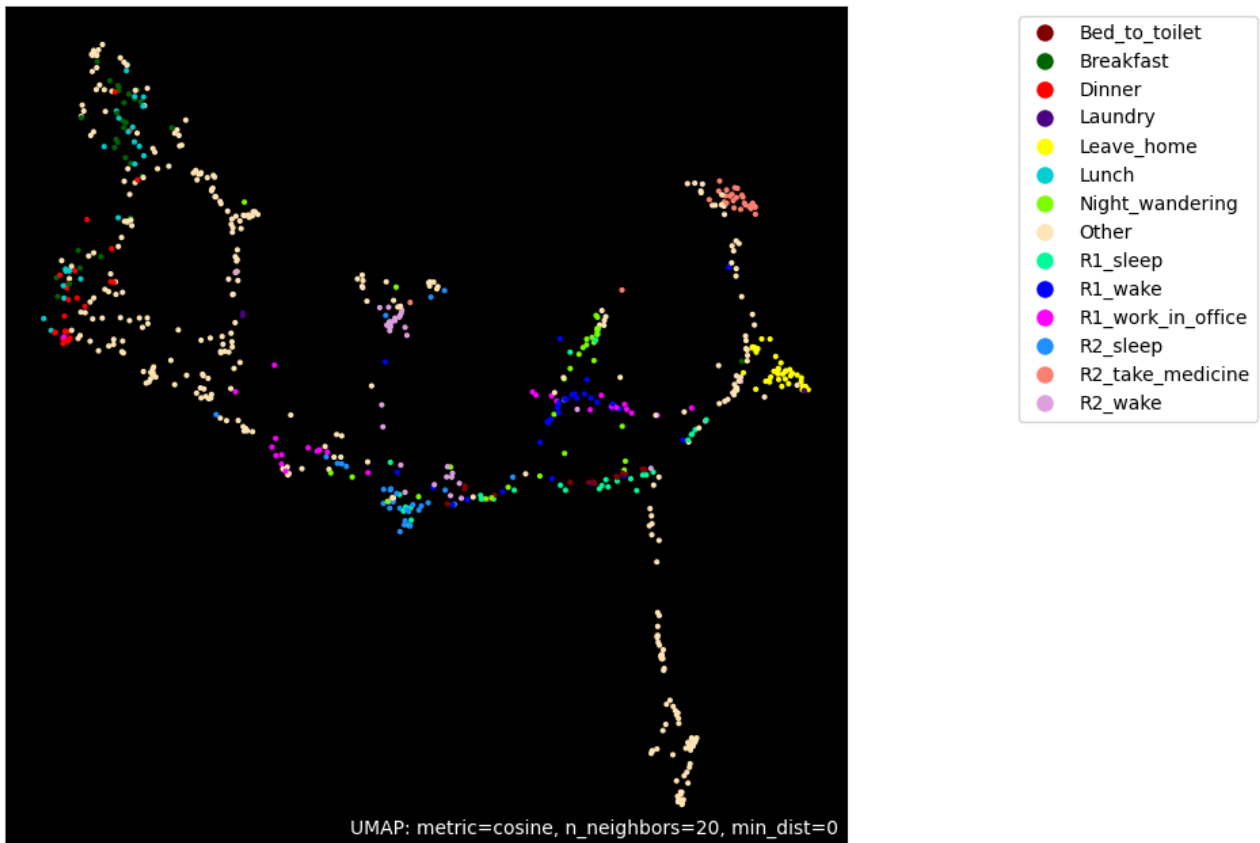
(a) ELMo



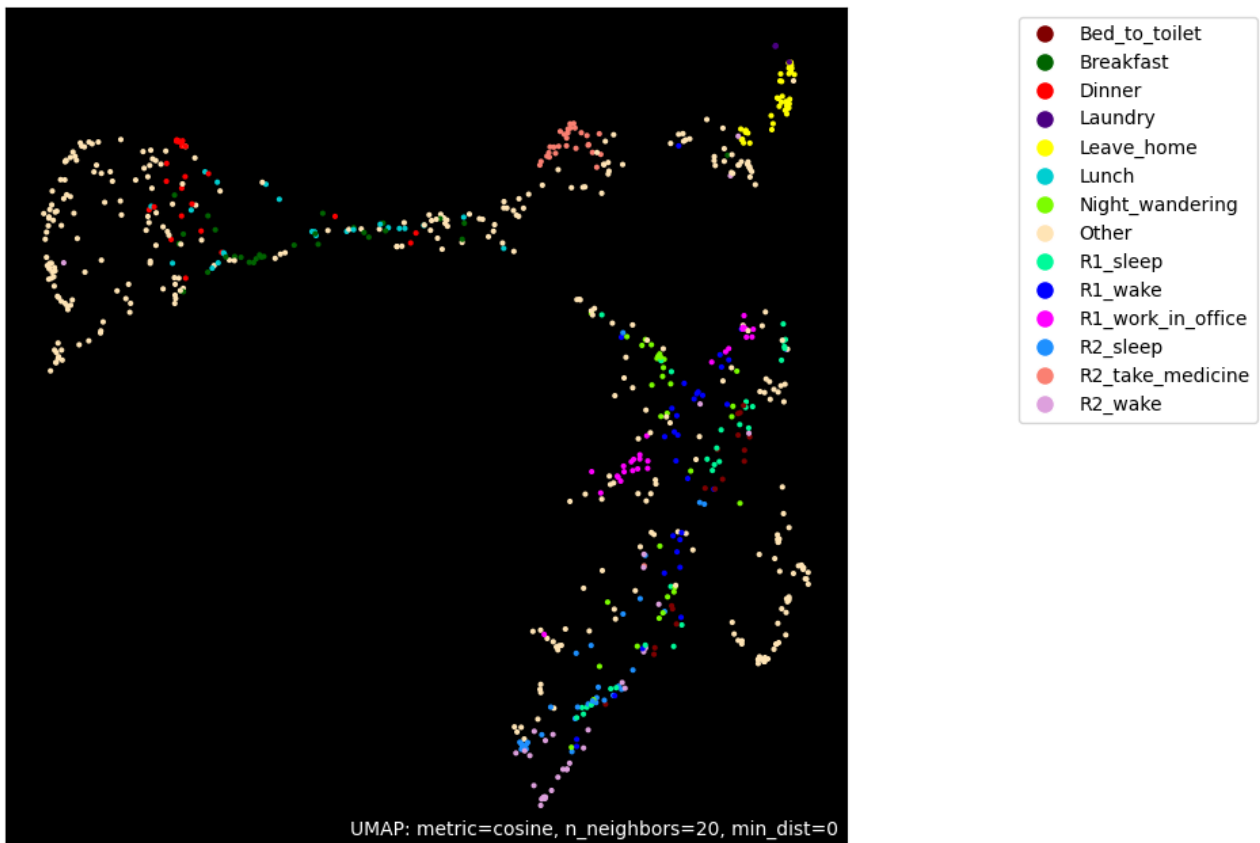
- Bed_to_Toilet
- Chores
- Desk_Activity
- Dining_Rm_Activity
- Eve_Meds
- Guest_Bathroom
- Kitchen_Activity
- Leave_Home
- Master_Bathroom
- Master_Bedroom_Activity
- Meditate
- Morning_Meds
- Other
- Read
- Sleep
- Watch_TV

(b) GPT

Figure 13: Visualization of Activity Embeddings for the Milan Dataset (Test Set) Generated by ELMo (Top) and GPT (Bottom)



(a) ELMo



(b) GPT

Figure 14: Visualization of Activity Embeddings for the Cairo Dataset (Test Set) Generated by ELMo (Top) and GPT (Bottom)